

ABNB: From Macro Signals to Management Guidance and Equity Reaction

An institutional research architecture, evidence audit, and reusable IC decision template

Evidence available through 03 September 2026 | System documentation, not a live forecast

IC VERDICT The governance and research process works. Forecast edge remains unproven; trading edge remains untested. Continue the build, but do not promote a production signal from the present retrospective evidence.

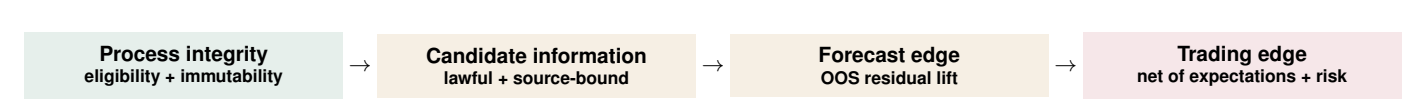
What we can claim – and what we cannot

Edge layer	Verdict	Evidence today	IC interpretation
Process edge	WORKS	Strict point-in-time controls, rejection logs, baselines, immutable packets, checksums, and resolution plumbing.	The process makes errors diagnosable and prevents hindsight from being mislabeled as evidence.
Candidate information edge	CANDIDATE	A revenue-weighted travel-activity composite is strongly associated with guidance growth in a retrospective slice.	Worth underwriting prospectively; it has 0 strict-PIT historical rows and is not yet a forecast feature.
Forecast edge	UNPROVEN	No stable walk-forward residual improvement over seasonal and prior-quarter baselines has been demonstrated.	Correlation with the raw seasonal guidance level is not sufficient.
Trading edge	UNTESTED	Direct guidance-to-return relationships are weak; comparable consensus, positioning, and implied-move history is absent.	Do not infer a stock signal from guidance direction or level alone.

Decision in one line

FUND	Point-in-time consensus history, management-policy features, and prospective travel-native data preservation.
TEST	The residual $G_t - \widehat{G}_t^{\text{outside}}$ in preregistered walk-forward folds, then compare it with consensus.
DO NOT TRADE YET	Raw guidance changes, retrospective snapshots, or the 0.784 diagnostic association.

The edge hierarchy



Scope discipline. This brief describes a reusable research and decision system. It does not manufacture a September 2026 forecast, claim proven alpha, or convert current snapshots into historical point-in-time observations.

Systematic macro-to-equity waterfall

A causal chain with explicit information clocks

The unit of work is an event packet: one target event, one offset-aware cutoff, one eligible evidence set, four linked forecasts, and a later immutable resolution.

0	Event clock	Fix event, target metric, guided period, unit, and <i>as_of_utc</i> . Evidence must have verified first availability strictly before cutoff; equality fails. Retain every rejection and reason.
1	Macro regime	Classify growth, inflation, FX, rates, employment, and consumer balance-sheet conditions using vintages actually observable at cutoff. State regime uncertainty, not a single narrative.
2	Travel system	Translate macro conditions into travel demand , carrier/room capacity , market pricing , and regulatory supply . Align geography, frequency, lag, and booking-versus-stay timing.
3	ABNB operating state	Nowcast nights and experiences booked, ADR, gross booking value, take rate, revenue, and margins. Produce an operating P10/P50/P90; do not jump directly from macro to guidance.
4	Management policy	Model how management maps the operating distribution into the formal range: conservatism, range width, seasonality, known headwinds, prior cadence, and source-bound language.
5	Market expectations	Freeze contemporaneous consensus, dispersion, bank expectations, implied move, and relevant positioning. A later-restated estimate is not point-in-time consensus.
6	Surprise vector	Construct joint actual and guidance surprises for comparable metrics and periods: $s_t = [s_t^{\text{actual}}, s_t^{\text{guide}}]$. Raw sequential guidance change is not a surprise.
7	Market state	Condition on valuation, factor exposures, crowding/positioning, implied volatility, macro regime, and recent price path. These variables mediate how the same surprise is priced.
8	Reaction distribution	Estimate downside/base/upside return distributions and decision thresholds, net of the implied move and benchmark. Keep the reaction forecast separate from the guidance forecast.
9	Resolve and learn	Attach reviewed actuals, issued guidance, consensus surprises, and price outcomes without changing the frozen forecast. Attribute error to economic, policy, expectations, or reaction layers.

Four linked forecasts – four distinct error terms

Forecast	Output	Primary diagnostic
Economic nowcast	Reported operating P10/P50/P90	Did macro/travel features predict ABNB operations?
Guidance policy	Formal guidance distribution and width	Did policy mapping add value beyond seasonality?
Expectations	Consensus and dispersion at cutoff	Was the estimated surprise defined correctly?
Reaction	Conditional stock-return distribution	Did surprise plus market state explain repricing?

Control principle. An LLM-generated number never substitutes for a failed deterministic calculation. Each number must be a cited input, computed result, explicit human assumption, or signed evidence-linked adjustment with a falsifier.

What already works

A functioning control plane, not yet a proven signal

23	20	PIT discipline	Immutable
preserved guidance events	numeric ranges	rejects retained	packet + checksums

Layer scorecard – reusable reporting interface

Question	Available evidence	Baseline	Candidate edge	Current status	Next gate
Is evidence eligible?	Availability, vintage, manifest, definition, period, unit, evidence ID; rejected rows retained.	Strict cutoff rule.	Faster lawful preservation with defensible timestamps.	WORKS	Prospectively lock native releases.
What is the operating state?	Guidance history; macro and travel hypothesis inventory; source-governed panel.	Seasonal and prior-quarter reference views.	Accommodation-native demand, geographic weights, capacity and price bridge.	PARTIAL	Build PIT operating P10/P50/P90.
What will management guide?	23 events, 20 numeric ranges, source-bound targets.	Seasonal naive, policy-adjusted level, median width, residual interval.	Historical policy language and signed adjustments.	MACHINERY	Populate eligible policy features; walk forward.
What did the market expect?	Interface exists; comparable PIT history does not.	Contemporaneous consensus midpoint and dispersion.	Estimate revisions and bank expectation distribution.	GAP	Acquire and freeze consensus history.
How will the stock react?	ABNB/SPY next-close event-return panel.	Zero/benchmark return; implied move when available.	Joint surprise vector conditioned on market state.	UNTESTED	Assemble surprise, position and valuation history.
Can learning be audited?	Frozen JSON, eligibility CSV, memo, checksums, parent links, resolution concepts.	Deterministic rehearsal.	Layer-specific error attribution and model registry.	WORKS	Resolve real packets prospectively.

Operational assets already in place

Event and target history	Twenty-three preserved guidance events span qualitative and numeric disclosures; the builder independently recounts the panel before publication.
Point-in-time governance	Reference period, observation, first availability, vintage, collection, and cutoff are distinct. Ambiguous timestamps and current-snapshot backfills are rejected, not silently dropped.
Baseline machinery	Seasonal naive, management-policy adjustment, median range width, and residual interval concepts exist before any feature promotion.
Packet integrity	New run directories fail closed on overwrite; updates require new IDs and parent links; checksums make mutation visible.
Market panel	ABNB and SPY event windows exist for all 23 events, enabling descriptive excess-return diagnostics.
Source governance	Provider permissions, collection path, evidence status, rejections, and stop reasons are preserved across hypothesis work.

What the rehearsal proves: workflow and control integrity. What it does not prove: forecast accuracy, backtest performance, or alpha. The fixture is synthetic and must remain labeled as such.

Evidence and stop matrix

Rejected, inconclusive, control-only, and prospective evidence stays visible

ID / sleeve	Economic mechanism	Historical evidence result	Status	Required next gate
H-001 H.10	Broad USD moves can affect reported growth and management framing.	Only strict-PIT historical signal. Direction: 9/16 overall; 3/4 early, weakening to 6/12 later. No walk-forward magnitude test.	WEAK / UNSTABLE	Preregistered folds versus seasonal and prior-quarter baselines; test residual magnitude and regimes.
H-002 ECB FX	EUR/USD translation and European demand proxy.	Economically coherent, but exact initial publication timestamps are unresolved; sensitivity timing cannot establish eligibility.	INELIGIBLE	Defensible historical release-time archive.
H-003 BLS lodging CPI	U.S. lodging pricing can inform ADR and substitution.	Historical release timestamps/vintages not independently verified; current API values cannot be backfilled.	INELIGIBLE	Source-bound historical vintage calendar.
H-004 WSF	Ferry activity may proxy a narrow travel corridor.	Automated path/permission gate failed; no lawful edge observations.	STOP	Explicit permission and decision-relevant breadth.
H-005 Orange County TDT	Local transient-tax receipts can proxy lodging spend.	Collection/vintage and narrow-geography limitations prevent a comparable history.	STOP	Lawful vintages, wider geography, and mapped lags.
H-006 NYC OSE	Enforcement can constrain legal short-term-rental supply.	Regulatory control only. The available vintage covers 4 later events—far below a predictive regime history.	CONTROL ONLY	Multiple comparable policy transitions and supply linkage.
U.S./Europe	Airport activity plus European platform nights can proxy demand.	Strong retrospective level relationship; all historical rows are current-snapshot diagnostics, not strict PIT. SFO is too narrow for the U.S. sleeve.	PROSPECTIVE	Archive native releases; broaden U.S. airport coverage.
50-source panel	Breadth can reveal robust cross-geography activity.	Valuable hypothesis inventory, but many Eurostat products are not independent providers and snapshots cannot be retroactively promoted.	INVENTORY	Predeclare features, vintages, redundancy and promotion gates.

How to read a stop

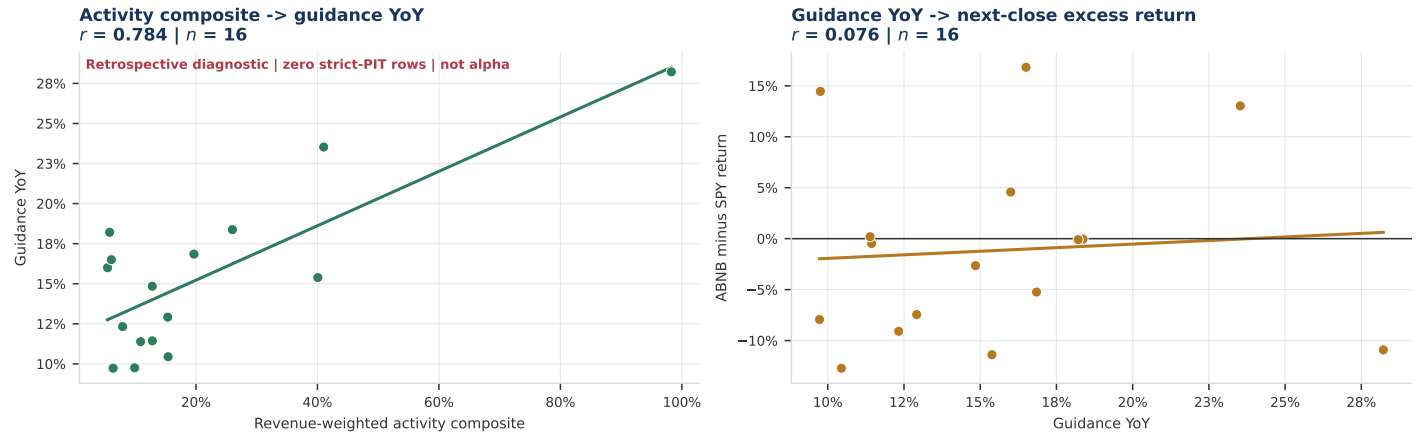
Rejected	Fails permission, timestamp, manifest, or cutoff.	Retain row plus exact rejection reason; exclude from modeling.
Inconclusive	Mechanism is plausible, evidence cannot decide.	Preserve hypothesis; do not score as a miss or a hit.
Control-only	Relevant to risk/regime but lacks predictive history.	Use as a qualitative constraint, never a trained feature.
Prospective	Can become eligible only after future releases are timestamped and archived.	Freeze now; evaluate only after enough out-of-sample events.

Non-negotiable: no present-day snapshot, reconstructed release time, or later-restated estimate may be represented as evidence available at a past forecast cutoff.

Correlation contrast

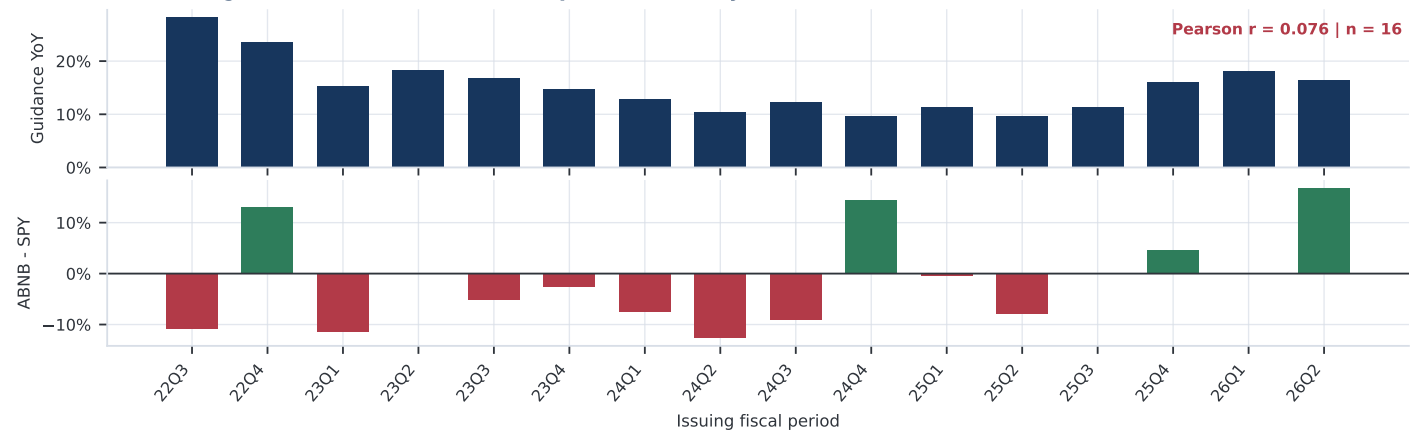
Strong activity-to-guidance association; weak direct guidance-to-return link

RETROSPECTIVE DIAGNOSTIC – ZERO STRICT-PIT ROWS – NOT ALPHA. Revenue-weighted activity composite versus guidance YoY: Pearson $r = 0.784$, Spearman $\rho = 0.388$, $n = 16$.

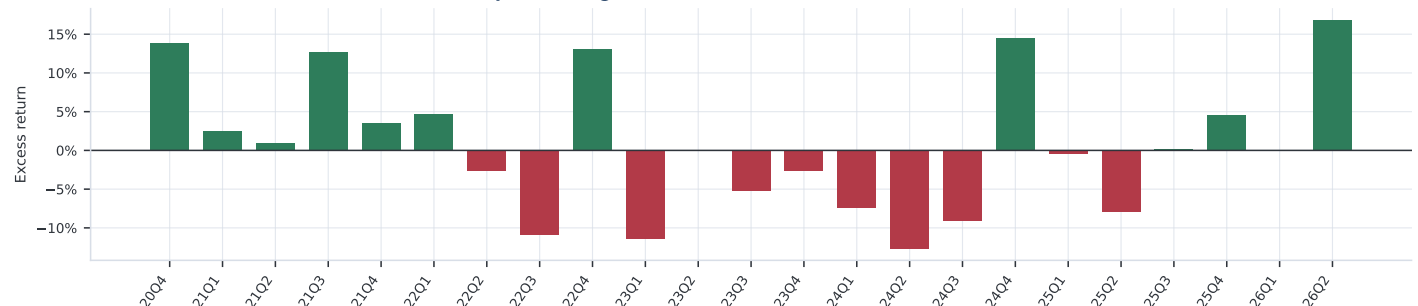


Diagnostic	Result	Sample	Interpretation / limitation
Revenue-weighted composite vs guidance YoY	$r = 0.784$; $\rho = 0.388$; $n = 16$		Strong level association, but only +0.008 Pearson above 50/50 and +0.003 above SFO; 0 strict-PIT rows.
Composite level vs guidance acceleration	$r = -0.627$; $\rho = -0.504$; $n = 15$		Negative relationship: the level fit does not transfer to changes in growth.
Acceleration direction concordance	53.3%	$n = 15$	Near coin-flip directional agreement.
Guidance YoY vs excess return	$r = 0.076$	$n = 16$	Weak direct mapping from guidance growth to ABNB minus SPY next-close return.
Guidance acceleration vs excess return	$r = 0.082$	$n = 15$	Also weak.
Sequential guidance pct. change vs excess return	$r = 0.032$	$n = 19$	Seasonal quarter mix dominates raw sequential changes. Direction aligns only 7/19 (36.8%).

Guidance growth and next-close stock performance by event



ABNB next-close excess return across 23 preserved guidance events



Return panel. Across 23 events, mean ABNB return is +1.16%; mean excess return is +0.71%; median excess return is -0.05%. These unconditional summaries do not establish tradability and are visibly dominated by event dispersion.

Where the investable edge must come from

Forecast the residual, define the surprise, then price the state

1. Remove the easy seasonal level

$$G_t = \hat{G}_t^{\text{outside}} + \Delta_t^{\text{policy}} + \varepsilon_t,$$

$$\text{research target } R_t^G = G_t - \hat{G}_t^{\text{outside}}.$$

The outside view includes same-guided-quarter seasonality, prior guidance, historical range width, eligible macro/travel inputs, and source-bound policy priors. Candidate features must improve the out-of-sample distribution of R_t^G , not merely track raw G_t .

2. Compare like-for-like with contemporaneous expectations

$$s_t^{\text{guide}} = \frac{\hat{G}_t - C_t^{\text{guide}}}{\text{scale}(C_t)}, \quad s_t^{\text{actual}} = \frac{A_t - C_t^{\text{actual}}}{\text{scale}(C_t)}.$$

Consensus C_t must match metric, guided period, currency basis, and publication cutoff. Estimate dispersion and revision path matter because identical midpoint surprises can carry different information content.

3. Price the joint surprise vector, not a single headline

$$\mathbf{s}_t = \begin{bmatrix} s_t^{\text{actual}} & s_t^{\text{guide}} \end{bmatrix}, \quad r_{t+1}^{\text{excess}} \sim f(\mathbf{s}_t, \text{positioning}_t, \text{implied move}_t, \text{valuation}_t, \text{macro regime}_t).$$

This structure permits beat/raise, beat/cut, miss/raise, and miss/cut states. It also acknowledges that a “good” guide can underperform when already priced or when valuation and macro beta dominate.

4. Keep decisions downstream of distributions

Decision input	Required representation	Kill condition
Operating state	P10/P50/P90 for nights, ADR, GBV, take rate, revenue, margins	No eligibility-complete bridge from evidence to operations
Guidance	P10/P50/P90, expected range width, policy bridge	No OOS residual lift over named baselines
Expectations	PIT midpoint, dispersion, revisions, comparability	Restated or post-cutoff consensus
Reaction	Conditional downside/base/upside and benchmark excess return	Missing comparable surprise and market-state history
Position	Direction, size, horizon, stop/falsifier, implied-move hurdle	Expected edge does not exceed uncertainty and execution costs

Raw guidance changes are seasonally distorted. They are useful descriptive fields, but they are not substitutes for consensus surprise and should not drive a position directly.

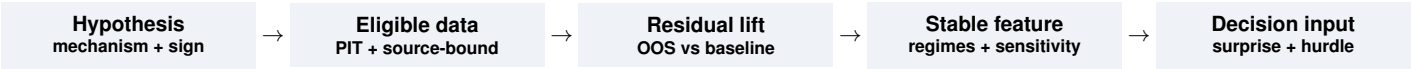
Core falsifier. If eligible features cannot produce stable out-of-sample residual improvement after matching seasons and regimes, the information-edge thesis fails even if in-sample level correlations remain high.

Ranked research roadmap

Sequence the missing evidence before fitting a reaction model

Rank	Workstream	Deliverable	Promotion gate
1	Build point-in-time consensus and estimate-dispersion history.	Timestamped metric/period-matched consensus snapshots, revision paths, and dispersion with provenance.	Comparable availability before every event cutoff; no later-restated values.
2	Populate source-bound management-policy features.	Structured features from eligible SEC exhibits, letters, and pre-cutoff transcripts: conservatism, cadence, range width, named headwinds, wording changes.	Definitions frozen before scoring; target-event leakage excluded.
3	Run the preregistered H.10 walk-forward test.	Fold-by-fold residual MAE/RMSE, calibration, sign hit rate, and regime stability versus seasonal and prior-quarter baselines.	Stable out-of-sample residual improvement; not a one-regime directional count.
4	Prospectively archive accommodation-native European releases.	Release-native files, exact first-availability timestamps, hashes, revisions, geographic mapping, and lags.	Enough eligible events to assess incremental residual value.
5	Broaden the U.S. airport sleeve beyond SFO.	Predeclared multi-airport demand/capacity composite with population, route, and geography coverage controls.	Improvement over simple/broad baselines survives leave-one-airport-out tests.
6	Fit a reaction model only after the upstream history exists.	Comparable actual/guidance surprises, consensus dispersion, implied move, valuation, positioning proxies, ABNB/SPY returns.	Frozen specification; honest test window; decision-relevant net edge after uncertainty.

Promotion ladder



Minimum evidence packet for every feature

Definition and mechanism	Time and provenance	Test and decision
Metric, unit, geography, expected sign, lag, frequency, coverage, transformations.	Reference period, first availability, revision/vintage, collection time, source ID, manifest status, checksum.	Named baseline, preregistered folds, missingness policy, stability checks, residual metric, promotion threshold, falsifier.

Research resource allocation

Highest value	Consensus history and management-policy features close the two largest missing links between operating evidence, guidance surprise, and equity reaction.
Medium value	Prospective travel-native preservation creates future eligibility but cannot repair the historic sample today.
Lowest value now	More undifferentiated scraping or a premature nonlinear reaction model increases feature count without solving vintage, comparability, or sample-size constraints.

Promote a feature only after stable out-of-sample residual improvement. A high contemporaneous correlation, a plausible mechanism, or a broad source panel is not a promotion gate.

Reusable pre-earnings IC template

One packet from event definition to go/no-go decision

Field	Required entry	Field	Required entry
Event	Issuing quarter, event UTC, guided period, target metric/unit	Cutoff	Offset-aware <i>as_of_utc</i> ; version, run ID, parent ID
Eligible evidence	Approved manifest IDs and accepted rows; rejected rows/reasons attached	Operating P10/50/90	Nights, ADR, GBV, take rate, revenue, margins; bridge from baseline
Guidance distribution	P10/P50/P90, expected low/high, range width, policy adjustments	Consensus	PIT midpoint, dispersion, revisions, comparability and source packet
Surprise vector	Actual and guide surprises for matched metrics/periods	Reaction distribution	Down/base/up excess return, horizon, benchmark, key conditionals
Implied move	Options-implied hurdle and timestamp	Position	Direction, size, entry, horizon, loss budget, no-trade zone
Risks	Data, model, policy, regime, crowding, valuation, execution	Falsifiers	Evidence or outcomes that invalidate the thesis before/after event

Go / no-go gates

Gate	Decision	Required evidence
Eligibility	GO / NO-GO	Every modeled row approved, fully defined, and first available strictly before cutoff; rejects retained.
Minimum sample	GO / HOLD	Preregistered minimum for the feature and test design is met; descriptive <i>n</i> alone does not count as test folds.
Baseline lift	GO / NO-GO	Stable OOS improvement in residual loss/calibration versus seasonal and prior-quarter baselines.
Stability	GO / HOLD	Sign, magnitude, and error distribution survive time/regime/geography sensitivity checks.
Decision relevance	TRADE / NO-TRADE	Surprise is material versus consensus dispersion and implied move after positioning, valuation, and risk.

Compact chronology and required stops

Phase A – readiness	Preserved the 23-event target panel, defined strict PIT fields, replayed H.10/ECB/BLS candidates, and separated eligibility from sensitivity assumptions.
E0 – discovery slate	Screened the initial multi-source hypothesis inventory; exact lawful collection paths did not produce promotable historical edge observations.
E1 – source tests	Retained source-level outcomes, rejections, and control-only results rather than forcing a winner.
Broad scrape	Expanded observation breadth under source governance; current snapshots remained prospective and could not be backfilled as PIT.
50-source expansion	Added many Eurostat product views; breadth increased, provider independence and vintage eligibility did not.
U.S./Europe comparison	Found the $r = 0.784$ level diagnostic, but 0 strict-PIT rows, weak acceleration behavior, and only marginal lift over simpler sleeves.
Stops enforced	Permission, publication-time, vintage, geographic breadth, and regime-history failures stopped promotion. OSE stayed a regulatory control.

Compact source notes

Public anchors: Airbnb guidance targets are source-bound to SEC exhibits and releases (<https://www.sec.gov/edgar/browse/?CIK=1559720>); Federal Reserve H.10 dated releases (<https://www.federalreserve.gov/releases/h10/>); Eurostat data browser (<https://ec.europa.eu/eurostat/databrowser/>); SFO Open Data (<https://data.sfgov.org/>).

Auditable local artifacts: `outputs/01a065f3-9f9d-7c33-a034-621059cc8c6f/ABNB_edge_guidance_stock_reaction.xlsx`; `outputs/01a065f3-9f9d-7c33-a034-621059cc8c6f/abnb_us_europe_guidance_panel.csv`; `research/readiness/20260903T053309Z_abnb_readiness/cohort_2020q4_2023q2_replay.csv`; `research/readiness/20260903T053309Z_abnb_readiness/cohort_2023q3_2026q2_replay.csv`. Displayed statistics, source hashes, and contract checks are emitted in `docs/forecasting/abnb_ic_brief/generated/metrics.json`.

IC output: an auditable probability distribution, an explicit comparison with expectations, and a position only if evidence, baseline, stability, and decision-relevance gates all pass.